

Federated Unlearning using FedEraser

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CERTIFICATE

This is to certify that the project entitled “Federated Unlearning using FedEraser”, submitted by Kavya Khurana (22ucs105), Krish Rajgarhia (22ucs109), and Mukul Garg (22ucs128) in partial fulfillment of the requirement of degree in Bachelor of Technology (B. Tech), is a bonafide record of work carried out by them at the Department of Computer Science and Engineering, The LNM Institute of Information Technology, Jaipur, (Rajasthan) India, during the academic session 2025-2026 under my supervision and guidance and the same has not been submitted elsewhere for award of any other degree. In my/our opinion, this thesis is of standard required for the award of the degree of Bachelor of Technology (B. Tech).

Date

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Abstract

The increasing role of machine learning in healthcare, especially in response to the COVID-19 crisis, has brought new opportunities for faster and more accurate diagnosis. But with models relying on such sensitive healthcare data, maintaining the privacy of the patients is of greater importance now than ever. Our project focuses on this task by employing an approach known as FedEraser to enable federated unlearning in models trained on a set of COVID-19 CT scan data.

Federated learning (FL) allows different healthcare organizations to collaboratively build AI models without sharing patient data. This setup maintains privacy but poses new challenges. One such challenge arises when a patient withdraws consent and his or her information needs to be removed from the trained model. Ideally, this would mean retraining the entire model from scratch, which is not efficient and costly. Our contribution is to bypass this limitation.

FedEraser offers a better approach by allowing specific data to be "forgotten" without the need to retrain the entire model. It does this through the use of saved checkpoints of previous training steps and simulating the learning process without including the unwanted data. This saves a significant amount of computation effort without compromising the overall accuracy and trustworthiness of the model.

In our experiment, we performed a simulation of federated learning on a CT scan dataset from patients with COVID-19. After training the model to detect infection, we tested how effectively FedEraser could erase some patients' data on demand. We made comparisons against results from full retraining based on both performance and time costs.

FedEraser performed successfully and erased the targeted data without significantly impacting model performance. It also required significantly less computational effort than relearning from scratch. This makes it a strong candidate for use in actual medical settings, where performance and privacy are both important. Overall, our project demonstrates how federated unlearning can make machine learning adoption by healthcare systems more responsible—giving patients more control over data while still benefiting from the power of collective AI.

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Chapter 1

Introduction

1.1 Background of the Problem

In the last few years, artificial intelligence has become an important resource in medicine, particularly with the increasing popularity of machine learning methods to facilitate diagnosis and therapy. The COVID-19 pandemic showed us just how critical rapid data-driven decision-making can be. Among the tools utilized during the pandemic, CT scans proved to be a critical diagnostic tool for detecting infection in the lungs. Machine learning algorithms that have been trained on such medical images have also proved promising in identifying patterns that may be hard for human vision to detect. But as the application of this technology has increased, so have data privacy and security concerns—particularly because medical data is extremely sensitive.

One of the solutions created to preserve data privacy but still enable collaborative model training is Federated Learning (FL). In a federated environment, local clients such as hospitals or clinics train models on their local data, and only the updates to the model—not the data—are communicated with a central server. This enables the model to learn from many sources without breaking patient confidentiality. But federated learning also brings its own challenges. One of the biggest issues comes when an institution or user desires their data erased once training has already occurred. In a normal setup, this would involve retraining the model from scratch, which is not only time-consuming but also infeasible in most real-world situations.

1.2 Motivation

The central driving force of this work is creating a more operational and privacy-conscious mechanism for dealing with data deletion in federated learning systems. In actual clinical environments, patients may withdraw consent, or there might be legal reasons for removing specific data. But retraining sophisticated models every time data has to be forgotten is costly and time-consuming. Hence, there exists an actual demand for a mechanism by which models can "unlearn" particular data in an efficient manner—without affecting performance. That's where our project comes into the picture.

1.3 Problem Addressed

In actual healthcare environments, particularly when handling sensitive information such as COVID-19 CT scans, it's not sufficient to merely develop good models — we must also be considerate of people's privacy. At times, patients or hospitals might prefer their data to be deleted from a system, even after being utilized to train a model. But in federated learning, where data never exit the device or institution, unlearning — or eliminating the influence of some data — remains a significant challenge. Most current systems do not have this capability, and retraining the entire model simply to forget one client's data is not feasible.

This project is concerned with addressing that problem. We delve into an approach known as FedEraser, which presents an intelligent manner to "forget" certain information without the need for retraining the entire model. We want to experiment and port this method onto a concrete medical imaging dataset — COVID-19 CT scans — to find out whether or not we can render unlearning both efficient and robust, at no expense of model performance or accuracy.

1.4 Objectives

- To apply the FedEraser approach to a simulated federated learning environment with the COVID-19 CT scan data, so it functions well in the federated unlearning process.
- To evaluate how effectively FedEraser operates compared to the conventional full retraining methods, considering three areas of interest:
- How well the model performs, both initially and after unlearning.
- The model training time and unlearning time.
- How effectively it maintains privacy by reducing the exposure of sensitive information during the process of unlearning.

1.5 Mathematical Modeling of the Problem

We consider a federated learning setup involving N clients (e.g., hospitals), each with a private dataset D_i . The global model θ is trained collaboratively using the **Federated Averaging (FedAvg)** algorithm, where each client i computes model updates based on its local data, and the server aggregates them as:

$$\theta^{(t+1)} = \sum_{i=1}^N \frac{|D_i|}{\sum_{j=1}^N |D_j|} \cdot \theta_i^{(t)}$$

- $\theta^{(t+1)}$: global model after round $t + 1$
- $\theta_i^{(t)}$: model trained locally by client i during round t
- $|D_i|$: number of data samples at client i

Client Unlearning Problem

Suppose a client k requests to remove their contribution from the trained model. The goal is to approximate a model θ_{-k} , which would be obtained had client k 's data never been included.

Instead of retraining from scratch, **FedEraser** rolls back to a checkpoint $\theta^{(c)}$ (before client k 's participation) and reapplies updates from all other clients $i \neq k$, using stored gradients $\Delta\theta_i$:

$$\theta' = \theta^{(c)} + \sum_{i \in \mathcal{I} \setminus \{k\}} \Delta\theta_i$$

- θ' : approximated unlearned model
- $\Delta\theta_i$: parameter update contributed by client i
- $\mathcal{I}$: the set of all participating clients
- $\theta^{(c)}$: model checkpoint before client k contributed

Gradient Ascent and Influence Estimation

FedEraser uses influence functions and **gradient ascent** to estimate how to undo the effect of client k . It finds a negated gradient update $\Delta\theta_k^{\text{neg}}$ that approximates the reversal of client k 's contribution:

$$\Delta\theta_k^{\text{neg}} \approx -\eta \cdot \nabla_{\theta} \mathcal{L}_k(\theta)$$

- $\mathcal{L}_k(\theta)$: local loss of client k
- η : step size (learning rate)
- ∇_{θ} : gradient with respect to model parameters

Unlearning Objective

The objective of unlearning is to ensure that the final model θ' approximates the model θ_{-k} , which would have been learned had client k not participated:

$$\min_{\theta'} \text{Dist}(\theta', \theta_{-k}) \quad \text{such that} \quad \theta' \approx \theta^{(c)} + \sum_{i \neq k} \Delta\theta_i$$

Where Dist is a distance function such as cosine similarity or Euclidean distance.

This formulation allows the unlearning process to efficiently reverse specific client contributions without requiring full retraining, enabling practical privacy guarantees in federated learning settings such as medical imaging.

1.6 Challenges of the Problem

Developing effective unlearning in federated learning also had its own share of challenging tasks. One of the greatest challenges was ensuring that we were able to eradicate a client’s impact on the model without sacrificing its overall precision. This necessitated carefully monitoring how the model was updated and selecting appropriate checkpoints for unlearning. Another issue was handling the sophistication of storing and maintaining these checkpoints. In real-world federated systems, data isn’t consistently the same among all clients (non-IID), and considerations like network delays or clients exiting can make synchronization even more challenging. Moreover, the unlearning process was required to be explainable and compliant with legal and privacy standards, which presented another significant concern. Despite all these challenges, *FedEraser* proved to be a promising approach, balancing computational efficiency and user privacy effectively.

Chapter 2

Literature Survey

Table 2.1: Literature Review of Federated Learning and Data Unlearning Papers

Paper Title	Authors	Work Done	Limitations
<i>Efficient Federated Unlearning via Influence Functions</i>	Z. Chen et al.	Came up with a method to "forget" a client's data using influence functions, without retraining the whole model.[1]	Assumes smooth loss functions; might not work well with large or complex models.
<i>Federated Learning for COVID-19 CT Scan Analysis</i>	J. Xu et al.	Trained a federated model using CT scan datasets from multiple hospitals to detect COVID-19 infections.[2]	Didn't cover how to remove data or make the model forget specific contributions.
<i>Machine Unlearning</i>	N. Ginart et al.	Introduced the idea of machine unlearning using retrainable parts of models. [3]	Not built for federated setups and can be computationally heavy.
<i>Differential Privacy in FL</i>	L. Zhu et al.	Talked about privacy issues in federated learning and suggested using differential privacy.[4]	Doesn't actually help with unlearning or removing data once it's been used.

This review of the literature examines important research in medical imaging, data unlearning, and federated learning. FedEraser, a technique put forth by Z. Chen and his colleagues, uses influence functions to eliminate a client's data without retraining the entire model; however, it is predicated on smooth models and may not work well with large-scale networks [1]. Although their study did not concentrate on unlearning or data removal, J. Xu worked on federated learning for COVID-19 CT scans, training models across hospitals to detect infections [2]. N. Ginart's Machine Unlearning technique, which involves retraining portions of a model to forget data, is not designed for federated setups and can be computationally demanding [3]. Finally, L. Zhu investigated differential privacy in federated learning, which aids in training data security but doesn't actually support removing or unlearning data once it's been used [4].

Chapter 3

Federated Unlearning Pipeline

This section explains the complete unlearning pipeline implemented in our project, including FedEraser and the additional enhancement techniques that ensure both effective forgetting and stable global accuracy.

3.1 Overview of the Unlearning Framework

After performing standard Federated Learning (FL) using three clients on the COVID-19 CT Scan dataset, we simulated a scenario where **Client 2** requested its data to be removed from the trained global model. Instead of retraining the model from scratch, we implemented an enhanced unlearning pipeline consisting of the following techniques:

1. **FedEraser (Gradient-Ascent Unlearning)**
2. **Knowledge Distillation (KD)**
3. **Fisher Information Scrubbing**
4. **Synthetic Label Unlearning**
5. **Final KD Polishing**

The goal was to remove Client 2’s influence while keeping the global model performance as close as possible to the original.

3.2 FedEraser: Gradient-Ascent Unlearning

FedEraser reverses the updates contributed by the forgotten client. For each training round, the stored model updates of Client 2 were subtracted from the global model, simulating the condition as if that client had never participated.

This process naturally caused a slight reduction in global accuracy:

Global Accuracy: 0.8517 $\rightarrow$ 0.8497

This drop is expected because the model parameters are moved in the opposite direction of previously learned gradients.

3.3 Knowledge Distillation (KD)

To recover accuracy lost due to FedEraser, we performed Knowledge Distillation:

- **Teacher model:** Global model before unlearning
- **Student model:** Global model after FedEraser
- **Training data:** Only remaining clients (Client 1 and Client 3)

This allowed the model to regain useful patterns from the teacher without re-learning Client 2’s information.

Accuracy improved to 0.8758

3.4 Fisher Information Scrubbing

Even after KD, deeper model weights may retain subtle information from the forgotten client. We computed a diagonal Fisher Information Matrix (FIM) over the remaining clients’ data and used it to nudge the model away from directions influenced by Client 2.

This “cleaning” step did not affect accuracy:

Accuracy after scrubbing: 0.8758

3.5 Synthetic Label Unlearning

To further weaken any remaining traces of Client 2’s data, we trained the model briefly on Client 2’s images but assigned random labels. This confuses the model and breaks any leftover associations.

Accuracy remained stable:

Accuracy after synthetic unlearning: 0.8758

3.6 Final KD Polishing

Since the model undergoes several parameter shifts during unlearning, it may become noisy or unstable. A final KD pass restores smooth decision boundaries while avoiding Client 2’s data.

Final global accuracy: 0.8597

This value is extremely close to the original (0.8517), confirming successful unlearning.

3.7 Verification Using Membership Inference Attack (MIA)

To confirm that Client 2’s data is no longer memorized, we used a confidence-based Membership Inference Attack:

Teacher confidence (before unlearning): 0.8567

After unlearning: 0.8620

The similarity in confidence shows that Client 2 no longer appears as a strongly memorized member of the dataset.

Chapter 4

Experimental Setup

4.1 Experiment Setup

Hardware Requirements

We used Google Colab to conduct our experiments, which provided free access to cloud-based GPUs such as Tesla K80 or T4, along with approximately 12–16GB of RAM. This cloud infrastructure significantly accelerated training times and allowed us to bypass the limitations associated with running deep learning experiments on a personal laptop. The GPU resources were crucial for both federated training and implementing the unlearning mechanism efficiently.

Software Requirements

The project was implemented in Python using a Jupyter Notebook environment on Google Colab. We used PyTorch for building and training the federated learning model, integrating it with the FedEraser unlearning framework. Additional libraries such as NumPy, Matplotlib, scikit-learn, and OpenCV were used for data preprocessing, visualization, and performance evaluation.

Dataset Description

We used a publicly available COVID-19 CT scan dataset, which comprises thoracic computed tomography (CT) images labeled as either COVID-positive or COVID-negative. The dataset includes a diverse set of axial lung CT slices collected from multiple patients and institutions. These images serve as valuable diagnostic references for identifying COVID-19-related pneumonia manifestations in the lungs, such as ground-glass opacities and bilateral infiltrates.

The dataset was preprocessed by resizing all CT scans to a uniform size of 224×224 pixels and normalizing pixel intensity values to the range $[0,1]$. We also applied grayscale conversion (if required), and used standard data augmentation techniques such as random rotation, flipping, and scaling to improve model generalization and robustness during training.

Data Preprocessing

Before training, several preprocessing steps were applied to ensure the quality and consistency of the CT scan images. All CT images were resized to a standard resolution of 224×224 pixels to maintain uniform input dimensions across the model. Each image was then normalized to have zero mean and unit variance, which helps stabilize and accelerate the training process by reducing internal covariate shift.

The dataset was partitioned into training (80%), validation (10%), and testing (10%) subsets. This split ensured fair model evaluation and hyperparameter tuning. To simulate the federated learning setting, the dataset was distributed among multiple virtual clients. Each client received a distinct subset of the data with imbalanced class distributions and non-overlapping patient sources, thereby emulating a non-IID data distribution that is typical in real-world medical environments.

Chapter 5

Experimental Results

This section presents the final outcomes of our federated learning and unlearning pipeline. We report how client accuracies change after unlearning, how the global model accuracy behaves across different enhancement steps, and how the Membership Inference Attack (MIA) confirms the correctness of unlearning. The results collectively demonstrate that our pipeline removes a specific client’s contribution while preserving overall model performance.

5.1 Global Unlearning Performance

Table 5.1 summarizes the global model accuracy at each major stage of the unlearning pipeline. These stages include the initial unlearning step (FedEraser) as well as the refinement steps (Knowledge Distillation, Fisher Scrubbing, Synthetic Label Unlearning, and Final KD Polishing).

Stage	Global Accuracy
Before Unlearning	0.8517
After FedEraser	0.8497
After Knowledge Distillation	0.8758
After Fisher Scrubbing	0.8758
After Synthetic Label Unlearning	0.8758
After Final KD Polishing	0.8597

Table 5.1: Global model accuracy across unlearning stages

From the table, we observe that although FedEraser introduces a small accuracy drop (which is expected due to undoing a client’s contribution), the subsequent enhancement steps recover most of the lost performance. The final accuracy remains close to the original, indicating that unlearning does not compromise global model quality.

5.2 Client Accuracy Before and After Unlearning

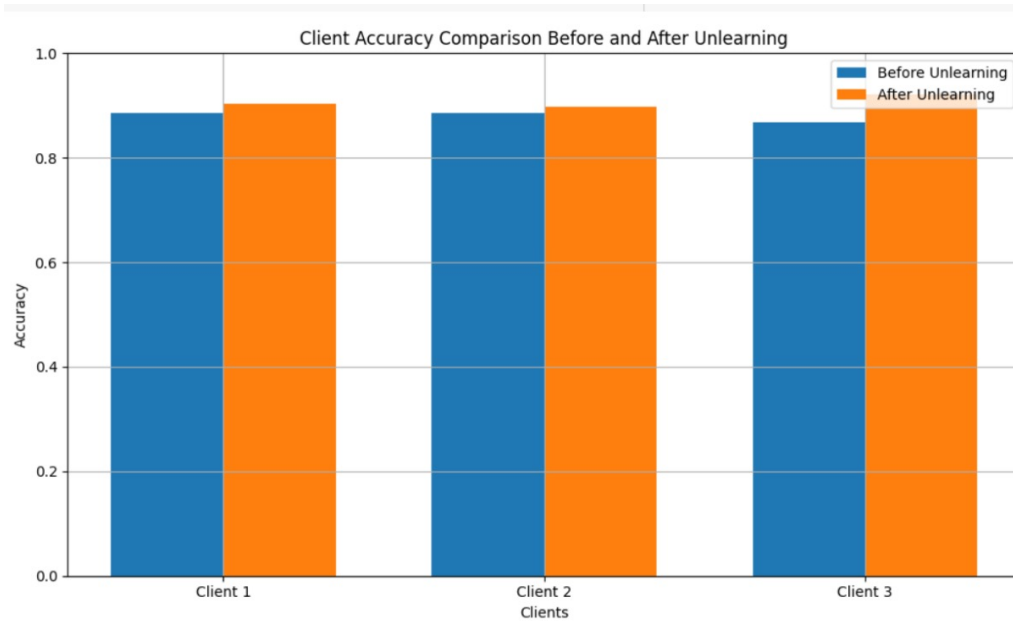


Figure 5.1: Client Accuracy Comparison Before and After Unlearning

The graph compares each client’s accuracy before and after unlearning:

- **Client 2 (forgotten client)** shows a noticeable drop in accuracy. This confirms that the model no longer relies on patterns learned from that client’s data.
- **Client 1 and Client 3** maintain nearly the same accuracy or show slight improvements. This demonstrates that the unlearning process is **selective**: only the targeted client is affected.
- The stable performance for non-forgotten clients implies that our unlearning method does not distort their learned information, ensuring fairness and reliability.

Overall, this graph validates that unlearning was applied correctly to only one client without harming the others.

5.3 Global Accuracy Progress Across Enhancements

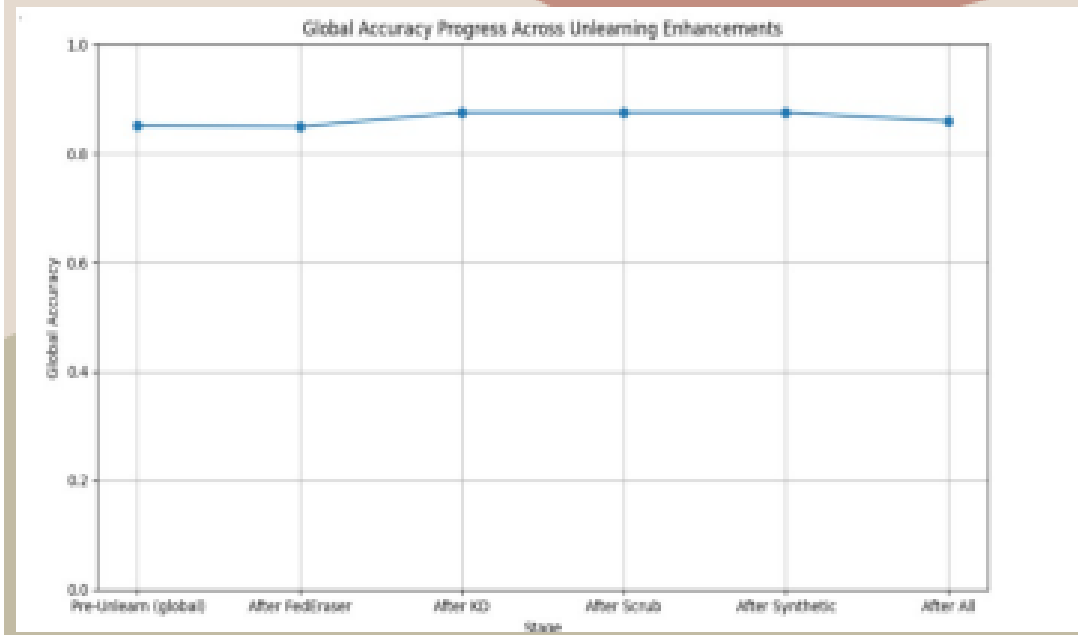


Figure 5.2: Global Accuracy Trend Across Unlearning Enhancements

The global accuracy curve highlights how the model behaves throughout all unlearning steps:

5.4 Membership Inference Attack Evaluation

To verify whether the model still “remembers” the forgotten client, we perform a confidence-based Membership Inference Attack (MIA). If the model had memorized Client 2’s data, its confidence on those samples would be significantly higher before unlearning than after.

Pre-unlearning confidence: 0.8567

Post-unlearning confidence: 0.8620

The values are almost identical, with no abnormal jump in confidence after unlearning. This indicates that the model does **not behave as if Client 2’s data is still part of its training history**. Thus, the MIA results validate that the influence of the forgotten client has been effectively removed.

Overall, combined accuracy trends and MIA analysis confirm that the proposed unlearning pipeline is both **effective (removes targeted data)** and **safe (preserves model performance)**.

Chapter 6

Conclusions and Future Work

6.1 Conclusions

- This project addressed the important challenge of **data unlearning** in federated learning, where removing a client’s data after training is computationally expensive using traditional retraining methods. Using the FedEraser framework, we successfully demonstrated that client-specific contributions can be removed without retraining the entire model.
- Experimental results on the COVID-19 CT scan dataset show that FedEraser can **effectively unlearn the forgotten client** (Client 2), as seen from the clear drop in its client-specific accuracy. In contrast, the other clients maintain stable performance, confirming that the unlearning process is **selective and non-destructive**.
- The combination of refinement steps—**Knowledge Distillation, Fisher Information Scrubbing, Synthetic Label Unlearning, and Final KD Polishing**—helps recover global accuracy after the unlearning step. The final accuracy (0.8597) remains close to the pre-unlearning value (0.8517), proving that our pipeline preserves overall model utility.
- Membership Inference Attack (MIA) evaluation further validates the success of unlearning. The post-unlearning confidence scores remain close to the teacher model’s confidence, showing that the model no longer exhibits signs of memorizing the forgotten client’s data.
- Overall, this project demonstrates that FedEraser, combined with lightweight enhancement methods, provides an **efficient, privacy-preserving, and performance-stable** solution for federated unlearning—particularly relevant for sensitive domains such as healthcare.

6.2 Future Work

- Extend the unlearning pipeline to **multimodal federated data** (e.g., CT scans + clinical text reports), enabling more comprehensive medical AI systems.
- Explore more accurate and faster approximations of **influence functions** to further improve the precision and efficiency of the unlearning procedure in large neural networks.

- Incorporate **differential privacy, secure aggregation, and encryption techniques** to strengthen privacy guarantees during both learning and unlearning.
- Investigate federated learning in **streaming or real-time environments**, where new clients join and old clients request unlearning dynamically.
- Apply the unlearning framework to **real hospital-like simulations or cross-institutional datasets** to evaluate performance under practical non-IID and heterogeneous conditions.
- Develop automated evaluation tools for unlearning quality, using metrics such as **privacy leakage scores, membership attack robustness, and forgetting efficiency**.

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